

their contributions to pathology of multiple type of neurodegenerative disease, including ALS, frontotemporal dementia (FTD), and Alzheimer's disease (AD), have been increasingly noted since the first observation that C9orf72 repeat expansions disrupt nuclear transport (Zhang et al., 2015). Cleveland's group found that aggregations of RNA binding proteins, nuclear pore proteins, and possibly other proteins could not be reversed by cycloheximide, which is known to inhibit SG formation. However, it is notable that Lloyd and colleagues found that aggregates of similar proteins that were induced by other stresses could be reversed by other SG inhibitors, such as Isrib (Zhang et al., 2018). This raises the possibility that even the cytoplasmic aggregates of nuclear pore proteins might exhibit crosstalk with the SG pathway.

These studies provide important advances in our knowledge of the mechanisms of formation of pathological aggregates (Figure 1). They highlight a pathological TDP-43 granule that is not a SG but, in some cases, evolves through a SG and in other cases evolves independently of SGs. Future studies will need to elucidate the relative proportion of

pathological TDP-43 aggregates that accumulate through each pathway in patients with ALS.

DECLARATION OF INTERESTS

Benjamin Wolozin is co-founder, chief scientific officer, and a member of the board of directors of Aquinnah Pharmaceuticals, Inc.

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Chandelier Cells Swipe Right for L1CAM

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Establishing a functional neuronal circuit requires not only synapsing with the right cell type, but also targeting the right subcellular compartment. In this issue of *Neuron*, Tai et al. (2019) identify the cell adhesion molecule L1CAM as integral to the mechanism by which chandelier cells establish subcellular compartment-specific innervation of pyramidal neurons in the mammalian cerebral cortex.

Meaningful connections are easier to form than ever before, as evidenced by the litany of apps now available to help one find a perfect match. If only it were so easy for neurons. Precise neural coupling relies not only on axonal navigation to a defined neuronal subtype located within

a particular region, but also on specific interactions between the presynaptic cell and a subcellular compartment of its target (Williams et al., 2010). GABAergic interneurons of the cerebral cortex display some of the best-characterized subcellular targeting in the mammalian central ner-

vous system. Many subtypes within this class each target a distinct subcellular region of recipient pyramidal cells (PyNs), multipolar projection neurons believed critical for advanced cognitive capability. The discriminating nature of GABAergic interneuron-PyN connectivity may reflect



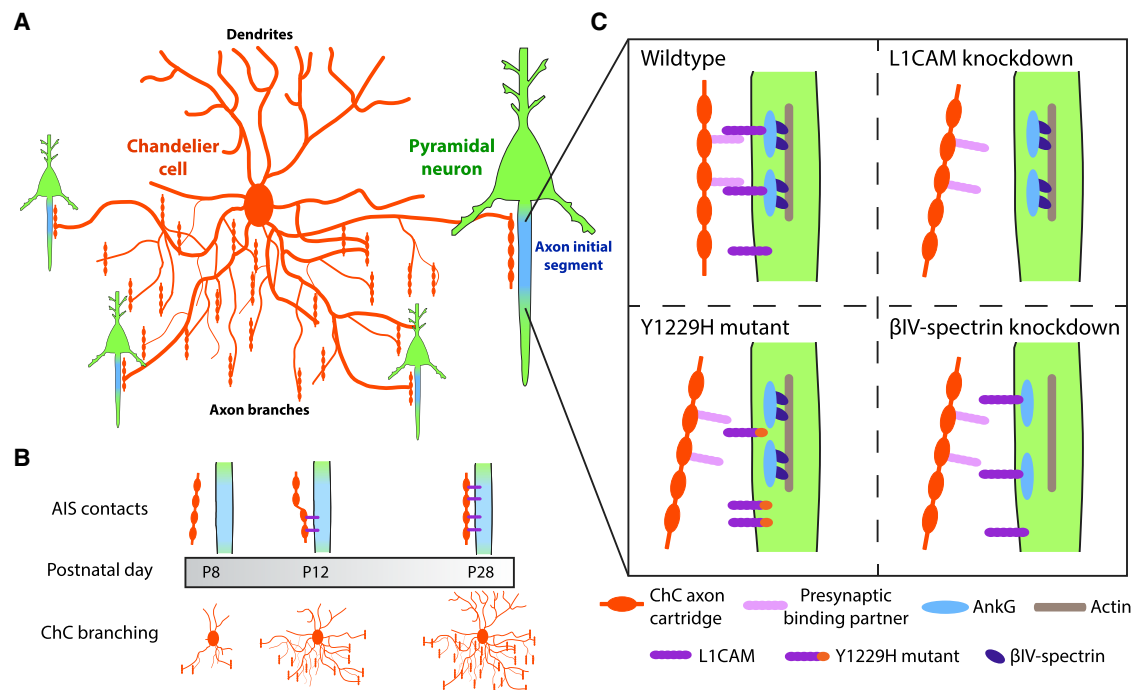


Figure 1. The Mechanistic Basis for Chandelier Cell Innervation of Pyramidal Neurons at the Axon Initial Segment

(A) A single chandelier cell (ChC) forms axo-axonic contacts with the axon initial segments (AISs) of multiple pyramidal neurons (PyNs).

(B) Establishment of ChC-PyN innervation coincides with increased ChC axonal branching and L1CAM expression during postnatal development.

(C) The L1CAM-ankyrinG-βIV-spectrin complex is critical for ChC-PyN contact. Contacts do not form if individual components are knocked down or mutated to disrupt interactions. The presynaptic binding partner of L1CAM remains to be determined.

the marked polarity and specialization of the PyNs themselves, which maintain anatomically and functionally distinct subcellular compartments, believed to confer greater computational power to individual neurons (Huang, 2006). By synapsing so selectively, GABAergic inputs exert influence over almost all PyN activity, including plasticity, integration, and population-level synchronicity (Somogyi et al., 1998).

How do neurons decide on an appropriate match? Developmentally, subcellular specificity is determined following genetically defined axon guidance but preceding activity-dependent plasticity, thus the mechanisms dictating synaptic localization could conceivably be innate, experience-related, or a combination of the two. Previous evidence favors the former (Di Cristo et al., 2004). Transmembrane cell adhesion molecules (CAMs), as well as some secreted proteins (Favuzzi et al., 2019; Williams et al., 2010), have emerged as prime candidates for directing compartment-specific synapse development, allowing connections to be stabilized where complementary adhesion

proteins are present. In the cerebellum, basket cell axons navigate to the axon initial segment (AIS), important for initiating action potentials, of Purkinje cells via interactions with neurofascin-186 (NF186), a member of the L1CAM family (Ango et al., 2004). Similar mechanisms are found in the spinal cord, where sensory afferent expression of NB2 and Caspr4 complement expression of L1CAMs CHL1 and NrCAM in GABApre interneurons, comprising a molecular recognition system essential for proprioceptive microcircuit assembly (Ashrafi et al., 2014).

Chandelier cells (ChCs), a sparse population of GABAergic interneurons in the mammalian cerebral cortex, exhibit highly branched axons that terminate in vertical arrays of boutons known as “cartridges,” through which they innervate hundreds of cortical PyN AISs (Figure 1A). ChC connectivity defects are associated with conditions such as schizophrenia and epilepsy, but the mechanisms regulating the subcellular localization of ChC innervation have been little examined. In this issue of *Neuron*, Tai et al. (2019) characterize how

L1CAM, identified by an RNAi screen of candidate CAMs, directs ChC contacts to the PyN AIS.

The screen, which included both attractive and repulsive molecules, was achieved through *in utero* electroporation (IUE) of vectors co-expressing EGFP and miR30-based short hairpin RNA (shRNA) against a given candidate, directed into nascent layer II/III PyNs. IUE was performed at embryonic day 15.5 (E15.5) in mice carrying a Cre-dependent TdTomato and a tamoxifen-inducible Cre recombinase knocked into the *Nkx2.1* locus. This Cre is expressed in developing ChCs, thus tamoxifen given to the dam at E18.5 will fluorescently label ChCs with TdTomato (Taniguchi et al., 2013). Brains collected at ChC maturity (P28) were then immunostained for ankyrin-G (AnkG), an AIS marker, to determine any change in the percentage of PyN AISs (EGFP+ and AnkG+) innervated by ChC axons (TdTomato+) resulting from each knockdown. This precise labeling and shRNA strategy was used throughout the study.

Knockdown of only one of the 14 molecules screened had a significant impact

on ChC innervation. L1CAM, a pan-axonally expressed CAM found throughout the nervous system, was revealed to be the culprit, in that its knockdown reduced ChC-PyN contacts at the AIS by almost 80%. Despite expectations that NF186, so important to innervation of cerebellar Purkinje cells at the AIS (Ango et al., 2004), would be involved, decreasing NF186 levels by neither shRNA nor CRISPR had any effect on ChC contacts. The L1CAM knockdown-induced lack of connectivity between ChCs and PyNs could be rescued by supplying a vector for human L1CAM (hL1CAM) alongside the shRNA vector. hL1CAM differs by three base pairs from the mouse variant within the shRNA-targeted region, rendering it shRNA resistant.

ChCs are GABAergic interneurons; therefore, decreasing the number of ChC presynaptic terminals should concurrently reduce the presence of markers for GABAergic synapses. Tai et al. (2019) found this to be the case, with the number of puncta immunostained for both VGAT (required for GABA uptake into presynaptic synaptic vesicles) and gephyrin (needed for anchoring postsynaptic GABA receptors) decreasing following L1CAM knockdown. Importantly, the presence of these GABAergic markers in other cellular domains, such as the soma, was unaffected by the reduction in L1CAM levels, showing that L1CAM is important only for AIS innervation.

With the “what” established, the question of “when” this CAM is important in development inevitably emerged (Figure 1B). Using multiple time points between P8 and P28, Tai et al. (2019) investigated ChC TdTomato overlap with AnkG and gephyrin in PyNs. They found almost no ChC synapses at P8 but maximal innervation by P28, completely overlapping with gephyrin in mature GABAergic synapses. There was a large spike in connectivity at P12, which coincided with a marked increase in the characteristic branching of ChC axons, suggesting a link between the two occurrences (Figure 1B). L1CAM expression also increased over this time frame, almost doubling between P10 and P13, thus numerous processes may be required simultaneously to enable the development of these synapses. This was supported by the reduced connectivity at P14 (immediately after this ostensibly crit-

ical period) following L1CAM knockdown, implying that the effect of L1CAM knockdown shown earlier was at least partly due to synapses never forming in the first place.

But does L1CAM have a dual role, in both development and maintenance of ChC-PyN synapses? Tai et al. (2019) went on to show that L1CAM is required to maintain ChC-PyN connections once established, demonstrating the importance of lifetime expression of L1CAM. To achieve this, the L1CAM RNAi construct was re-engineered to render it tamoxifen inducible in order to provide tight temporal control over its expression. This resulted in an incompatibility with the previously implemented Nkx2.1-CreER line, so an alternative approach was used. In an impressive technical feat, IUE was done twice to target two distinct cell populations. The ventral medial ganglionic eminence, whence ChCs are derived, was first targeted at E13.5 with a vector for TdTomato. Two days later, two vectors were directed to the ventricular zone for PyN expression. One vector contained a Cre-dependent L1CAM shRNA (and EGFP), and the other a tamoxifen-inducible Cre recombinase. These three vectors combined therefore allowed for temporally controlled expression of the shRNA via tamoxifen, as well as visualization of ChC fibers on PyN cells. Tamoxifen administered at P28, restricting L1CAM knockdown to adulthood, resulted in fewer ChC-PyN contacts, albeit to a lesser extent than when it is knocked down earlier in development, implying a role for L1CAM in synaptic maintenance.

Finally, Tai et al. (2019) turned their attention to how a pan-axonally expressed adhesion molecule directs synaptic connections to the AIS, but not to other regions of the axon. The answer was found in the expression of AnkG, heretofore used as an AIS marker but which is capable of anchoring transmembrane proteins such as L1CAM to the cytoskeleton through its interaction with β IV-spectrin (Figure 1C) (Leterrier, 2016). L1CAM therefore exists in two forms: one that is bound to AnkG and one that is AnkG-free. In a twist on the elegant rescue experiment performed earlier, the group employed a form of (RNAi-resistant) human L1CAM (Y1229H), which harbors a mutation in its cytoplasmic

domain that reduces AnkG interaction. Resoundingly, this mutant was unable to rescue L1CAM knockdown as the wild-type human form had done previously (Figure 1C). Simply knocking down β IV-spectrin was similarly found to reduce ChC contacts at the PyN AIS (Figure 1C). These experiments therefore provide strong evidence that the L1CAM-AnkG- β IV-spectrin complex in the AIS is essential to correct ChC-PyN innervation. The function of the endogenous pool of diffusible AnkG-free L1CAM present in the distal portion of the axon remains an open question.

The investigations by Tai et al. (2019) have shed light on the mechanisms dictating the establishment and maintenance of highly specific neuronal connectivity and have primed the field with a host of exciting avenues still waiting to be explored. What is the consequence and significance for cellular and behavioral outputs should interneurons find themselves synapsing with the wrong region? Is the L1CAM-AnkG- β IV-spectrin complex sufficient to determine subcellular specificity, along with its as yet unidentified presynaptic binding partner, or are repulsive factors on other cells or subcellular compartments also important for accurate targeting of an axon's final destination? How prevalent is subcellular localization of synapses: are all synapses constrained by regionality or is it only important to the function of some connections? While much remains to be discovered in the field, revealing the mechanisms behind the specificity of cortical circuitry will undoubtedly aid us in understanding the fine-tuning of neurons, circuits, and the behavior of complex systems.

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The Yin and Yang of Arnt2 in Activity-Dependent Transcription

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Spatiotemporal regulation of neuronal gene expression is essential for proper functioning of neuronal circuits. In this issue of *Neuron*, Sharma et al. (2019) discover a dual role for Arnt2-NcoR2 protein complexes in the activity-dependent regulation of neuronal transcriptomes.

The rapid modification of cellular transcriptomes represents a fundamental mechanism for neuronal homeostasis and plasticity. During development, activity-dependent gene expression programs coordinate the functional wiring of neuronal microcircuits. In the mature brain, activity-dependent programs enable homeostatic scaling, for example, by elevating neuronal inhibition in cells exposed to excess excitation (Turrigiano, 2008). Lastly, in learning and memory, activity-dependent gene transcription might support the strengthening of connections between neurons that are active during encoding, thereby creating ensembles with spatiotemporal activity patterns that can be recreated during memory retrieval (Josselyn et al., 2015). Over the past decades, there has been major progress on the identification of gene regulatory elements and transcription factors driving the onset of activity-dependent transcripts in neurons. However, comparably little is known about mechanisms that repress target gene transcription prior to stimulation. Tight repression would minimize noise from

background expression and thus maximize the dynamic range that can be accomplished in response to stimulation. But how can robust repression be achieved without compromising the dynamics of gene induction in response to stimulation? Conceivably, a rapid onset of activation is critical to associate transcriptional responses with a particular set of stimuli. In this issue of *Neuron*, Sharma and colleagues (Sharma et al., 2019) now uncover transcription factor complexes that facilitate rapid switching from repression to activation of transcription in rodent neurons.

The spatiotemporal control of transcription across the entire genome is coordinated by a complex interplay of transcription factors, their binding sites in promoters and enhancers, and the accessibility of these binding sites. Long-term silencing of regulatory elements is frequently directed by DNA methylation, which alters chromatin structure and thus reduces access for transcriptional regulators (Lister et al., 2013). However, in the mature brain, there are alternative mechanisms for silencing that are more rapidly

reversible upon neuronal stimulation. At activity-dependent promoters, proteins that modify histone composition or histone modifications have important functions in silencing transcription prior to stimulation. For example, the nucleosome remodeler complex NuRD triggers inactivation of promoters (Yang et al., 2016) or recruitment of the histone deacetylase HDAC4 represses genes in inactive neurons (Sando et al., 2012). Nonetheless, the spatiotemporal regulation of the full repertoire of activity-dependent gene regulatory elements in neuronal cells—in particular at the level of enhancers—is poorly understood.

Sharma and colleagues (Sharma et al., 2019) now uncover novel mechanisms underlying the switch from repression prior to neuronal activity to gene activation with sensory stimulation in mouse neurons (Figure 1). They focus on the transcription factor Npas4, a neuron-specific, activity-induced immediate early gene, which binds promoters and enhancers and regulates transcription of late response genes in a cell-type-specific manner (Spiegel et al., 2014). For example, an Npas4-dependent